Software Design

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Introduction to Software Design

Objectives

- To explain what design is and how various types of design deal with different aspects of a product
- To present design as problem solving and outline the roles of abstraction and modeling in design
- To place design in the software life cycle
- To survey software engineering design methods

What is Software Design?

- Software design is the process of defining **how** a system will be structured and implemented **before** writing code. It's about making intentional choices rather than letting architecture emerge by accident.
- **Key distinction**:
- **Architecture**: High-level structure, major components, their interactions
- **Design**: Detailed structure within components, class relationships, algorithms
- **Implementation**: Actual code that realizes the design

Foundational Principles

- **SOLID Principles (Object-Oriented Design)**
- **Single Responsibility**: A class should have only one reason to change
- **Open-Closed**: Open for extension, closed for modification
- **Liskov Substitution**: Subtypes must be substitutable for their base types
- **Interface Segregation**: Many client-specific interfaces are better than one general-purpose interface
- **Dependency Inversion**: Depend on abstractions, not concretions

DRY (Don't Repeat Yourself)

- Each piece of knowledge should have a single, unambiguous representation

But be careful: Don't over-abstract too early

KISS (Keep It Simple, Stupid)

- Simplicity should be a key goal in design
- Complexity increases maintenance costs exponentially

YAGNI (You Ain't Gonna Need It)

- Don't add functionality until it's necessary
- Premature generalization is worse than missing an abstraction
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Design Process & Techniques

- **Design Process Steps**
- Requirements → High-Level Design → Detailed Design → Implementation → Refinement

2. Key Design Activities

Decomposition

- Break system into manageable modules/components

Cohesion: How closely related elements within a module are

Coupling: How much modules depend on each other (aim for low coupling)

Abstraction

- Hide complex details behind simple interfaces

Example: Database abstraction layer hides SQL differences

- **Design Patterns (Common Solutions to Recurring Problems)**
- **Creational Patterns:**
- **Factory:** Create objects without specifying exact class
- **Singleton:** Ensure only one instance exists
- **Builder:** Construct complex objects step by step
- **Structural Patterns:**
- **Adapter:** Make incompatible interfaces work together
- **Decorator:** Add functionality dynamically
- **Facade:** Provide simplified interface to complex subsystem

- **Behavioral Patterns:**
- **Strategy:** Define family of algorithms, make them interchangeable
- **Observer:** One-to-many dependency between objects
- **Command:** Encapsulate request as object

3. Modeling Techniques

- **UML Basics (Visual Communication)**
- Class diagrams (structure)
- Sequence diagrams (interactions)
- State diagrams (behavior)
- **Remember**: UML is a communication tool, not documentation
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Importance of Software Design

- We live in a designed world.
- Design is economically important and effects our quality of life.
- Software is becoming ubiquitous.
- The quality of software design has important consequences that software designers should be aware of and take seriously.

Software Products

A **software product** is an entity comprised of one or more programs, data, and supporting materials and services that satisfies client needs and desires either as an independent artifact or as essential ingredient in some other artifact.

Software Design Defined

Software designers do what designers in other disciplines do, except they do it for software products.

Software design is the activity of specifying the mature and composition of software products that satisfy client needs and desire, subject to constraints.

Design as Problem Solving

- An especially fruitful way to think about design is as problem solving.
- Advantages
  - Suggests partitioning information between problem and solution
  - Emphasizes that there may be more than one good solution (design)
  - Suggests techniques such as changing the problem, trial and error, brainstorming, etc.

Abstraction

Abstraction is an important problem-solving technique, especially in software design.

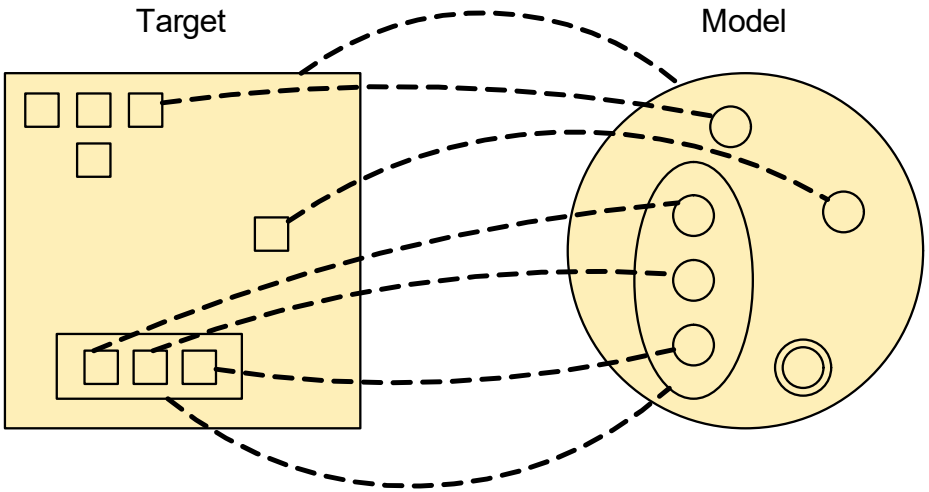
Abstraction is suppressing or ignoring some properties of objects, events, or situations in favor of others.

Importance of Abstraction

- Problem simplification
  - Abstracting allows us to focus on the most important aspects of a problem in (partially) solving it.
- Structuring problem solving
  - **Top-down strategy**: Solve an abstract version of the problem, then add details (**refinement**)
  - **Bottom-up strategy**: Solve parts of a problem and connect them for a complete solution

Modeling

A model represents a target by having model parts corresponding to target parts, with relationships between model parts corresponding to relationships between target parts.



Modeling in Design

- Modeling is used for the following purposes:
  - Problem understanding
  - Design creation
  - Design investigation
  - Documentation
- Modeling work because models abstract details of the target.
- Models can fail if important and relevant details are left out.

Static and Dynamic Models

A **static model** represents aspects of programs that do not change during program execution.

A **dynamic model** represents what happens during program execution.

- Static model examples include class and object models.
- Dynamic model examples include state diagrams and sequence diagrams.

Product vs. Engineering Design

- **Product designers** are concerned with styling and aesthetics, function and usability, manufacturability and manageability.
  - Industrial designers, (building) architects, interior designers, graphic designers, etc.
- **Engineering designers** are concerned with technical mechanisms and workings.
  - Structural, mechanical, chemical, and electrical engineers
- Design teams often include both product and engineering designers.

Software Product Design

Software product design is the activity of specifying software product features, capabilities, and interfaces to satisfy client needs and desires.

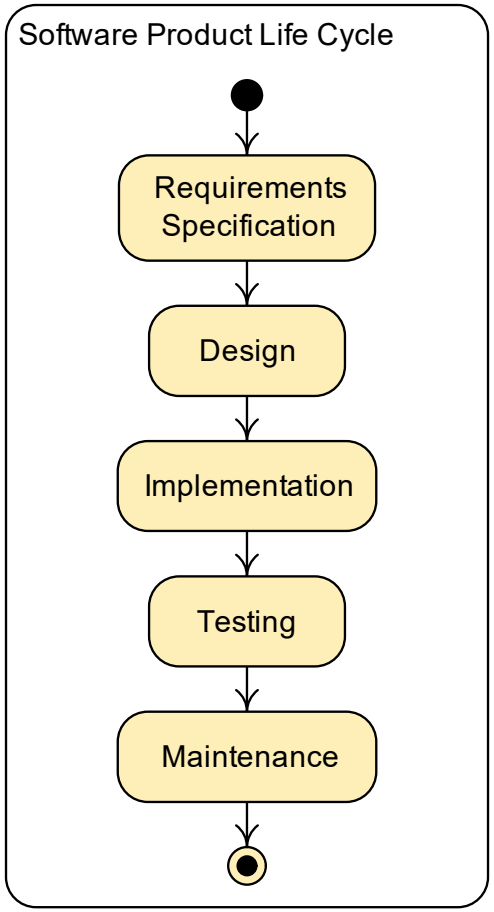
Requires skills in user interface and interaction design, communications, industrial design, and marketing

Software Engineering Design

Software engineering design is the activity of specifying programs and sub-systems, and their constituent parts and workings, to meet software product specifications.

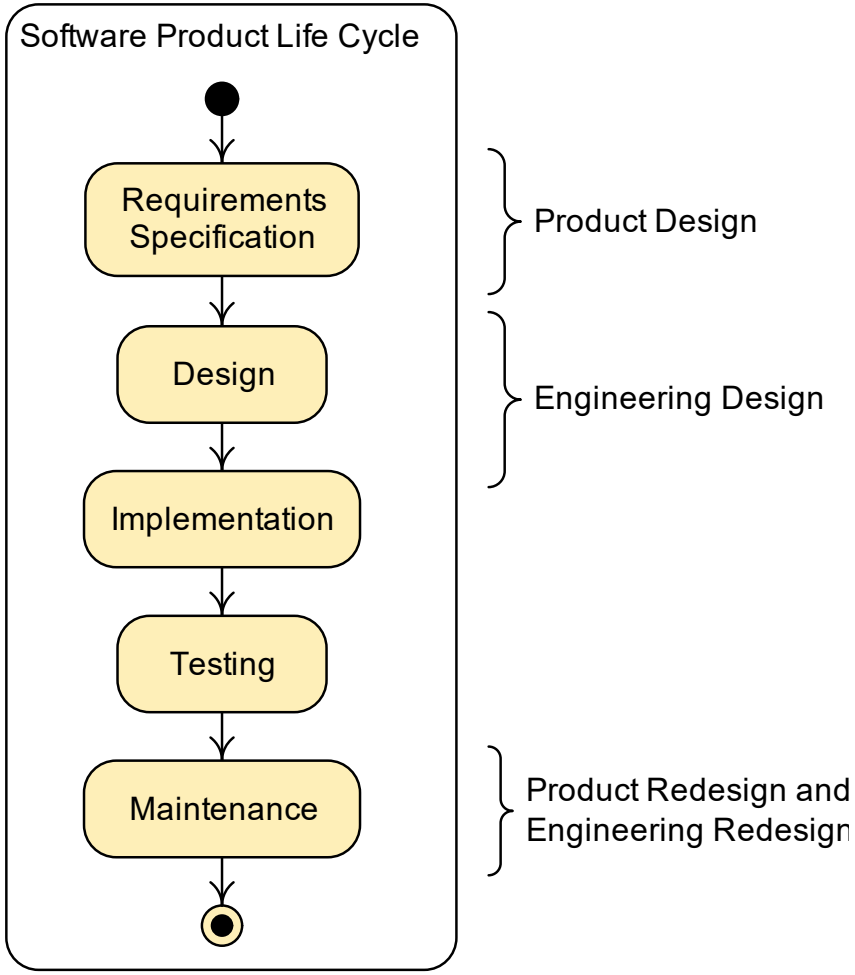
Requires skills in programming, algorithms, data structures, software design principles, practices, processes, techniques, architectures, and patterns

Waterfall Life Cycle Model

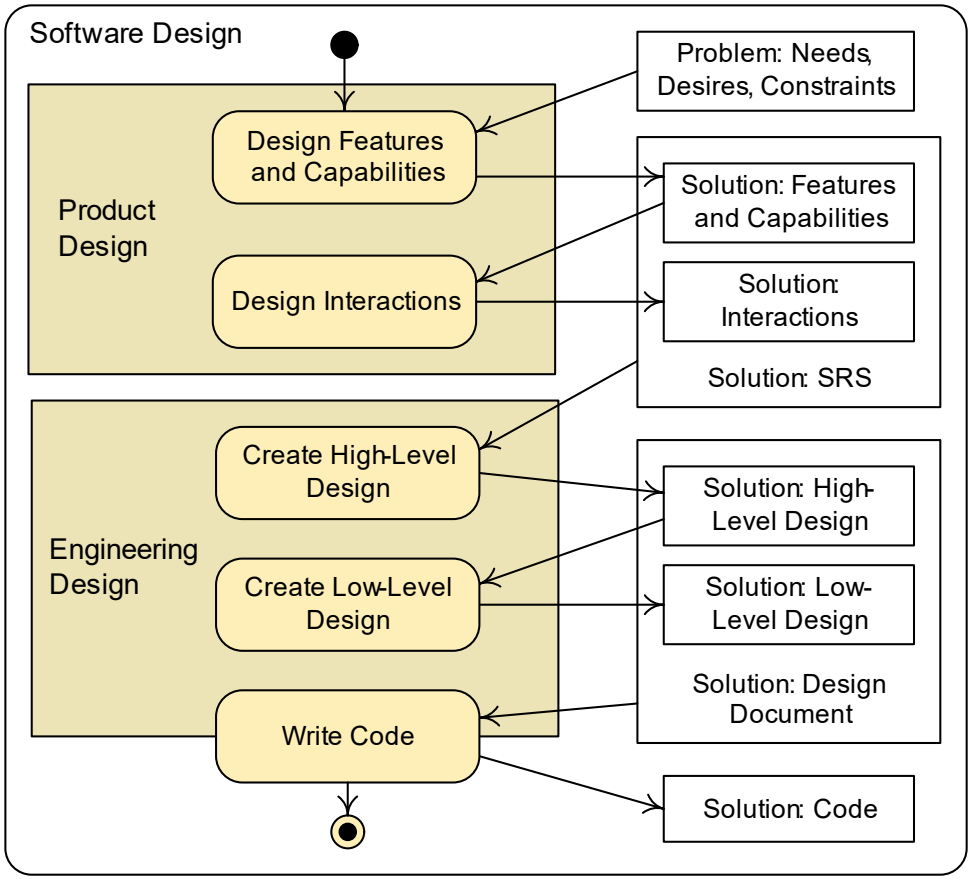


The waterfall model captures the logical, but not the temporal, relationships between software development activities.

Design Across the Life Cycle



Design Problems and Solutions



Software Design Method

A **software design method** is an orderly procedure for generating a precise and complete software design solution that meets clients needs and constraints.

Design Method Components

- *Design Process*—A collection of related tasks that transforms a set of inputs into a set of outputs
- *Design Notations*—A symbolic representational system
- *Design Heuristics*—Rules providing guidance, but no guarantee, for achieving some end
- Design methods also use **design principles** stating characteristics of design that make them better or worse.

END